Jyothis Chandran

Final-year PhD candidate · CMB component separation

Instituto de Física de Cantabria (CSIC–UC)
Santander, Spain

chandran@ifca.unican.es · jyothischandran1597@gmail.com
ORCID 0000-0002-5805-3901 · NASA ADS library · Google Scholar · jyothischandran.github.io · GitHub

Research profile

I work on component separation for CMB experiments, using needlet-based blind and semi-blind methods: NILC, GNILC, constrained ILC and multi-Stokes Hybrid NILC. I apply these to reconstruct the thermal Sunyaev–Zeldovich Compton-y signal and to suppress Galactic and extragalactic foregrounds. I am lead author of the current *Planck* PR4 y-map and of its polarization-aided, CIB-deprojected successor. Both are public, through Zenodo and the *Planck* Legacy Archive, and are used by the CMB and LSS communities. My current work extends these methods to the extraction of challenging Galactic foregrounds such as free-free emission, and to machine-learning approaches for point-source detection in CMB data.

Education

Jun 2022 – exp. Mar 2027
Santander, Spain

PhD in Science and Technology (Astrophysics and Particle Physics)
Universidad de Cantabria / Instituto de Física de Cantabria (IFCA)
Thesis: *Teasing new cosmological observables out of future CMB and radio surveys*
Supervisors: Dr R. Belén Barreiro and Dr Mathieu Remazeilles
Expected submission: 28 March 2027

2020 – 2021
Vitória, Brazil

Provisional enrolment, PPGCOSMO doctoral programme in Cosmology
Universidade Federal do Espírito Santo (UFES)
Admitted under Dr Oliver F. Piattella and completed graduate coursework remotely. Enrolment could not be taken up, owing to pandemic-related visa delays and a subsequent change in supervisory arrangements.

2015 – 2020
Thiruvananthapuram, India

BS-MS Dual Degree in Physics (Physics major, Mathematics minor)
Indian Institute of Science Education and Research (IISER-TVM)
CGPA 8.92/10. Thesis: *Harmonic-domain Internal Linear Combination for CMB analysis and the effect of biases* (Dr Soumen Basak).

Publications

2 first-author refereed papers · 2 conference proceedings · 4 LiteBIRD collaboration papers · 38 citations · NASA ADS, August 2026.

First-author refereed papers

P1 **J. Chandran**, M. Remazeilles, R. B. Barreiro. *Enhanced foreground mitigation in thermal SZ Compton-y maps via polarization and deprojection.* JCAP **08**, 062 (2026). doi:10.1088/1475-7516/2026/08/062, arXiv:2603.17799.

P2 **J. Chandran**, M. Remazeilles, R. B. Barreiro. *An improved Compton parameter map of the thermal Sunyaev–Zeldovich effect from Planck PR4 data.* MNRAS **526**, 5682–5698 (2023). doi:10.1093/mnras/stad3156. (37 citations)

Conference proceedings

C1 **J. Chandran**. *Enhanced foreground mitigation in thermal SZ Compton-y maps via polarization.* Proceedings of the 60th Rencontres de Moriond (Cosmology, 2026), accepted.

C2 **J. Chandran**, M. Remazeilles, R. B. Barreiro. *An updated and improved thermal SZ y-map from Planck PR4 data.* EPJ Web Conf. **293**, 00010 (2024), *mm Universe 2023*.

LiteBIRD collaboration publications

L1 K. Aizawa *et al. LiteBIRD mission overview after mission reformation.* arXiv:2607.25285 (2026).

L2 S. Vinzl *et al. Galactic science with the LiteBIRD satellite: spectral characterization of diffuse Galactic polarized emission at the angular power spectrum level.* arXiv:2607.20080 (2026).

L3 A. Novelli *et al. A cross-calibration approach for polarisation-sensitive detectors in CMB experiments: application to LiteBIRD's polarisation angle calibration.* arXiv:2606.23207 (2026).

L4 N. Raffuzzi *et al. Mitigating half-wave plate systematics at the map-making level: calibration requirements for LiteBIRD.* arXiv:2602.02410 (2026).

Public data releases

D1 **J. Chandran**, M. Remazeilles, R. B. Barreiro. *CIB-deprojected Compton y-parameter map of the thermal SZ effect from Planck PR4 data.* Zenodo (2026). doi:10.5281/zenodo.18876142
All-sky Hybrid NILC y-maps with selective CIB moment deprojection. 96 downloads in the first five months.

D2 **J. Chandran**, M. Remazeilles, R. B. Barreiro. *Compton y-parameter map of the thermal SZ effect from Planck PR4 data.*
Released through the **ESA *Planck* Legacy Archive**, ESA's official repository for *Planck* data products, and through Zenodo (2023), doi:10.5281/zenodo.7940376
615 downloads and 1,119 record views on Zenodo to date; in use by the wider tSZ and LSS communities.

Research projects & funding

2022 – present
Spanish national research projects, MCIN / AEI
Research carried out within PID2022-139223OB-C21, PID2022-140670NA-I00 and PID2025-173927NB-I00, all three of which have supported my conference travel. Additional support through the IFCA structural research programme of the Gobierno de Cantabria.

2024 – 2026
RadioForegroundsPlus (Horizon Europe) (HORIZON-CL4-2023-SPACE-01, GA 101135036)
Coordinator: Dr J. A. Rubiño-Martín (IAC); IFCA node led by Dr R. Belén Barreiro, with Dr Mathieu Remazeilles as project manager. Contributed to Galactic foreground mitigation in Compton-y maps using polarization as an external tracer (multi-Stokes Hybrid NILC; JCAP 2026); to full-sky free-free reconstruction with blind and semi-blind ILC methods; and to CNN-based point-source detection in total intensity.

2022 – 2027
"Concepción Arenal" Predoctoral Contract
Competitive doctoral contract awarded by the Universidad de Cantabria.

2015 – 2020
INSPIRE Fellowship, Ministry of Science and Technology, India
National merit fellowship for study and research in the basic sciences.

Dec 2018
CSIR-UGC NET (Physical Sciences): All India Rank 20

2018
IUCAA Visiting Students Programme award
Inter-University Centre for Astronomy and Astrophysics, Pune.

Talks & posters

Oct 2026
Tenerife, Spain
Hybrid TQU Needlet ILC to mitigate Galactic foregrounds in tSZ reconstruction (contributed talk, forthcoming)
Galactic Science and CMB Foregrounds, Instituto de Astrofísica de Canarias.

Sep 2026
Stockholm, Sweden
Improved foreground cleaning via polarization-aided cleaning and CIB deprojection in thermal SZ Compton-y maps (talk)
Cosmoglobe 2026: Global End-to-End Analysis at All Frequencies, Nordita. Conference week of a three-week programme.

Aug 2026
Leiden, Netherlands
Enhanced foreground mitigation in thermal SZ Compton-y maps via polarization-aided cleaning and CIB deprojection (talk)
COSMO-26: International Conference on Particle Physics and Cosmology.

May 2026
Enhanced foreground mitigation in thermal SZ Compton-y maps via polarization and deprojection (talk)
Simons Observatory Foregrounds Working Group meeting.

Mar 2026
La Thuile, Italy
Enhanced foreground mitigation in thermal SZ Compton-y maps via polarization (poster; proceedings accepted)
60th Rencontres de Moriond, Cosmology 2026.

Nov 2023
Tenerife, Spain

A refined Planck thermal SZ Compton y-parameter all-sky map to probe the entire hot gas in the Universe (talk)
IX Meeting on Fundamental Cosmology, Instituto de Astrofísica de Canarias.

Jun 2023
Grenoble, France

An updated and improved thermal SZ y-map from Planck NPIPE data (talk)
mm Universe 2023: Observing the Universe at millimetre wavelengths, LPSC.

Collaborations

Jan 2024 – present

Member, LiteBIRD Collaboration
Thermal SZ Project Study Group (project: *Mapping the hot gas in the Universe*), led by M. Remazeilles and G. Luzzi. Developing the component-separation pipeline for LiteBIRD forecasts on the relativistic SZ effect.

Earlier research experience

2020 – 2021

Harmonic ILC for multi-component reconstruction
With Dr Soumen Basak (IISER-TVM) and Dr Jacques Delabrouille (APC, Paris). Extended harmonic ILC towards the simultaneous reconstruction of several components; implemented and tested the pipeline on simulated LiteBIRD skies. Methodological groundwork for the ILC techniques used in my thesis.

May – Jun 2019
Heidelberg, Germany

Visiting researcher, Institut für Theoretische Physik
Heidelberg University, with Prof. Luca Amendola and Dr Oliver F. Piattella
Tested a model-independent expression for the anisotropic stress (gravitational slip) parameter in Horndeski gravity, comparing the analytic approximations of the Bellini–Sawicki parameterisation against numerical solutions across a range of scales. Built the Python analysis pipeline for the CLASS / hi_class output and took part in the cosmology group's weekly seminar programme.

May – Jun 2018
Pune, India

IUCAA Visiting Students Programme
With Dr Sanjit Mitra. Cross-correlation statistics of noise in laser interferometric gravitational-wave detectors.

Technical skills

Programming	Python (numpy, scipy, healpy, astropy, matplotlib), IDL/GDL, MATLAB, Mathematica, LaTeX, shell.
Cosmology software	HEALPix, NaMaster / pymaster, CAMB, CLASS / hi_class, Planck Sky Model (PSM), PySM, TensorFlow.
Methods	Needlet and harmonic ILC, GNILC, constrained ILC and moment deprojection, multi-Stokes Hybrid NILC, tSZ and CIB component separation, pseudo-C_ℓ power spectrum estimation, point-source masking and machine-learning point-source detection, one-point and higher-order map statistics.
Computing	Altamira (IFCA-CSIC, Spanish Supercomputing Network) and NERSC (active account); Slurm, Git, Linux.
Languages	English (TOEFL 112/120), Malayalam (native), Hindi, Spanish.

Outreach

2019

IISER-TVM schools outreach programme
Interactive session on radiation pressure and electromagnetic waves for secondary students, Evans Higher Secondary School, Parassala, India.

References

Dr Mathieu Remazeilles
CSIC Tenured Scientist
Instituto de Física de Cantabria
remazeilles@ifca.unican.es
PhD supervisor

Dr R. Belén Barreiro
Research Scientist
Instituto de Física de Cantabria
barreiro@ifca.unican.es
PhD supervisor

Dr Soumen Basak
Associate Professor
IISER Thiruvananthapuram
sbasak@iisertvm.ac.in
MSc supervisor